GYAN Analytics

Mathematical Foundations & Algorithmic Formulations

Platform: GYAN Analytics Enterprise BI

Target Architecture: Dual Hybrid ML + SLM + Domain RAG

Engineering Scope: Anomaly, Cluster, Correlation, RAG & QLoRA

Version: 2.0 (Production)

1. System Architecture & Mathematical Duality

GYAN Analytics is engineered on a **strict mathematical decoupling principle**: probabilistic neural generators (LLMs) are never permitted to calculate arithmetic sums, statistical distributions, or anomaly flags. Instead, high-performance deterministic algorithms execute on CPU, constructing a verified numerical foundation that is then synthesized with Domain Knowledge RAG playbooks and a fine-tuned Small Language Model (SLM) to produce consulting-grade intelligence.

Subsystem Layer	Core Mathematical Formulation	Algorithmic Implementation	Operational Output
Anomaly Isolation	Random subspace cuts, path length s(x, n)	Isolation Forest (100 iTrees)	Outlier index set, anomaly score, leakage triage
Bivariate Screening	Pearson covariance normalized by variance products	Sample Bivariate Covariance	Ranked driver pairs, R^2 determination, t-test
Cohort Partitioning	Within-cluster sum of squares minimization (WCSS)	Lloyd-Forgy k-Means	k centroid coordinates, cohort relative mass (%)
Data Governance	Bounded piecewise penalty objective H(D)	Deterministic Scorer	Data health score (0-100), traffic light rating
Knowledge RAG	Hilbert inner product in ■³■■ Euclidean space	all-MiniLM-L6-v2 + ChromaDB	Industry KPI benchmarks, profit matrices, ROI
Neural Synthesis	Low-Rank Adapter W = W■ + (α/r)BA, CLM Loss	Fine-Tuned Qwen2.5-0.5B (QLoRA)	Domain executive narrative, structured actions

2. High-Dimensional Anomaly Isolation (Isolation Forest)

Multivariate anomalies are isolated across continuous numerical feature spaces using an ensemble of 100 randomized binary isolation trees (iTrees). Outlier instances isolate near the root with few partitions:

$$s(x, n) = 2^{(- E(h(x)) / c(n))}$$

Equation 1: Normalized Isolation Forest Anomaly Scoring Function

Where **h(x)** is the tree path length for observation **x**, **E(h(x))** is its expectation across all 100 trees, and **c(n)** is the theoretical average path length of an unsuccessful search in a Binary Search Tree (BST):

$$c(n) = 2\ ln(n - 1) + 2\gamma - (2(n - 1) / n)$$

Equation 2: Expected BST Path Length with Euler-Mascheroni Constant ($\gamma \approx 0.5772156649$)

- **E(h(x)) → 0 ==> s(x, n) → 1:** Observation isolates near the root; definitive anomaly.
- **E(h(x)) → n - 1 ==> s(x, n) → 0:** Observation resides deep in partitions; regular instance.
- **E(h(x)) → c(n) ==> s(x, n) → 0.5:** Observation matches random uniform expectation.

Contamination Thresholding: The top quantile $\alpha \in [0.01, 0.05]$ defines decision threshold τ. Observations with $s(x, n) > \tau$ are flagged and forwarded to Phase 1 Triage.

3. Bivariate Covariance & Driver Significance

Key operational drivers and linear couplings are discovered via the Pearson Product-Moment Correlation Coefficient:

$$r_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\left[\sqrt{\sum (x_i - \bar{x})^2} * \sqrt{\sum (y_i - \bar{y})^2} \right]}$$

Equation 3: Pearson Bivariate Correlation Coefficient ($r_{xy} \in [-1.0, +1.0]$)

Coefficient of Determination (R^2): Quantifies explained variance: $R^2 = (r_{xy})^2$. Statistical hypothesis significance is verified via Student's t-transformation with $df = n - 2$:

$$t = r * \sqrt{\frac{(n - 2)}{(1 - r^2)}} \sim t_{(n - 2)}$$

Equation 4: Student's t-Distribution Test Statistic for Bivariate Significance

4. Unsupervised Cohort Partitioning (k-Means)

To group entities into natural behavioral cohorts without labels, continuous features are standardized via Z-score $z_{ij} = (x_{ij} - \mu_j) / \sigma_j$ and partitioned by minimizing Within-Cluster Sum of Squares (Inertia J):

$$J(C_1, \ldots, C_k) = \sum_{j=1}^{k} \sum_{x_i \in C_j} \| x_i - \mu_j \|^2$$

Equation 5: k-Means Inertia Objective Minimization Function

- **Assignment Step:** Assign each point to its nearest centroid manifold: $C_j^{(t)} = \{ x_i : \|x_i - \mu_j^{(t)}\|^2 \le \|x_i - \mu_l^{(t)}\|^2, \forall l \}$
- **Update Step:** Recalculate centroids as manifold mean: $\mu_j^{(t+1)} = (1 / |C_j|) \sum_{x_i \in C_j} x_i$

Dynamic Cluster Selection: Cluster count is dynamically bounded: $k = \min(k_{max}, \max(2, \lfloor N / 10 \rfloor))$, ensuring dense statistical representation for every cohort.

5. Data Governance & Reliability Scoring

Data quality is evaluated via a bounded deterministic objective function $H(D) \in [0, 100]$:

$$H(D) = \max(0, \min(100, 100 - P_{missing} - P_{duplicate} - P_{sparsity}))$$

Equation 6: Data Health Index Governance Function

Penalty Dimension	Mathematical Formulation	Cap Ceiling	Diagnostic Rationale
Missing Value Loss	$\lfloor (N_{missing} / (N \times D)) \times 100 \times 1.5 \rfloor$	35 points	Penalizes data incompleteness and unpopulated records.
Duplicate Row Loss	$\lfloor (N_{duplicates} / N) \times 50 \rfloor$	20 points	Penalizes redundant observations skewing sample variance.
Sparsity Penalty	$5 \times \sum \mathbb{1}(\text{NullRatio}(C_j) > 0.80)$	Unbounded	Deducts 5 pts per column containing >80% missing values.

6. Semantic Vector Embeddings & Dense Retrieval (RAG)

The Domain RAG layer maps user telemetry into a 384-dimensional dense semantic vector space via **all-MiniLM-L6-v2** ONNX models. Playbook chunks are ranked via Normalized Inner Product (Cosine Similarity):

$$\text{Sim}(q, d) = \cos(\theta) = \frac{q \cdot d}{\|q\|_2 \times \|d\|_2} = \sum_{i=1}^{384} q_i \times d_i$$

Equation 7: Normalized Cosine Similarity in $\mathbb{R}^{384}$ Semantic Vector Space

For unit-normalized vectors, cosine similarity maps directly to Euclidean distance: $\|q - d\|_2^2 = 2(1 - \cos(\theta))$, enabling sub-15ms local vector lookups across industry playbooks.

7. SLM Fine-Tuning & Parameter-Efficient Adaptation (QLoRA)

The generative executive narrative engine is powered by a fine-tuned **Qwen2.5-0.5B-Instruct** model. Low-Rank Adaptation (LoRA) freezes the base model weights $W_0 \in \mathbb{R}^{(d \times k)}$ and trains low-rank decomposition matrices:

$$W = W_0 + \Delta W = W_0 + (\alpha / r) \times B \times A$$

Equation 8: Low-Rank Adapter Decomposition (Rank $r = 16$, Scaling $\alpha = 32$)

Where $B \in \mathbb{R}^{(d \times 16)}$ is zero-initialized and $A \in \mathbb{R}^{(16 \times k)}$ is Gaussian-initialized. The scaling factor $\alpha / r = 32 / 16 = 2.0$ stabilizes gradient propagation. Fine-tuning optimizes the Autoregressive Causal Language Modeling (CLM) token-level cross-entropy loss:

$$\blacksquare_{CLM}(\theta) = -\frac{1}{T}\sum_{t=1}^{T} \log P(w_t \mid w_{<t}; \theta)$$

Equation 9: Autoregressive Token Cross-Entropy Optimization Objective

Optimization Schedule: Trained with AdamW optimizer using a Cosine Annealing learning rate schedule decaying from $\eta_{max} = 2 \times 10^{-4}$:

$$\eta_t = \eta_{min} + 0.5 * (\eta_{max} - \eta_{min}) * \left[1 + \cos\left(\frac{t}{T_{max}} * \pi\right) \right]$$

Equation 10: Half-Cycle Cosine Annealing Learning Rate Schedule

8. Algorithmic Financial Impact & Decision Matrix

Statistical telemetry is converted into prescriptive executive changes using a 3-phase corporate alpha matrix:

Operational Phase	Mathematical Trigger	Corporate Action Lever	Projected Financial Impact
Phase 1: Immediate Triage (Days 1-7)	Anomaly Mass: $s(x) > \tau$ (Outliers)	Cap discount bounds, enforce minimum basket thresholds.	**+3.5% to +6.0%** Gross Margin Recovery on high-velocity items.
Phase 2: Driver Scaling (Days 8-21)	Max Pearson coupling: $\max(\lvert r_{xy} \rvert)$ with $R^2 > 0.50$	Reallocate marketing spend and inventory to leading driver.	**+18% to +25%** Customer Lifetime Value (LTV); 30% lower churn.
Phase 3: Cohort Monetization (Days 22-30)	Centroid coordinates μ_j and share $\rho(C_j)$	Cohort retention paths; dynamic deadstock markdowns.	Releases **8% to 12%** trapped working capital; cuts holding costs by 15%.

Conclusion: By mathematically decoupling statistical computation (Isolation Forest, Pearson, k-Means) from probabilistic text generation (QLoRA, ChromaDB RAG), GYAN Analytics provides deterministic, mathematically verified enterprise business intelligence with zero numerical hallucination.